

Using Extreme Value Theory for Vehicle Level Safety Validation and Implications for Autonomous Vehicles

Daniel Åsljung^{ip}, Jonas Nilsson^{ip}, and Jonas Fredriksson^{ip}

Abstract—Much effort is put right now into how to make autonomous vehicles as capable as possible in order to be able to replace humans as drivers. Less focus is put into how to ensure that this transition happens in a safe way that we can put trust in. The verification of the extreme dependability requirements connected to safety is expected to be one of the largest challenges to overcome in the commercialization of autonomous vehicles. Using traditional statistical methods to validate complete vehicle safety would require the vehicle to cover extreme distances to show that collisions occur rare enough. However, recent research has shown the possibility of using near-collisions in order to estimate the frequency of actual collisions using Extreme Value Theory. To use this method, there is a need for a measure related to the closeness of a collision. This paper shows that the choice of this threat measure has a significant impact on the inferences drawn from the data. With the right measure, this method can be used to validate the safety of a vehicle. This, while keeping the validity high and the data required lower than the state of the art statistical methods.

Index Terms—Autonomous vehicles, error probability, road transportation, statistical analysis, vehicle safety.

I. INTRODUCTION

AUTONOMOUS vehicles are expected to contribute to a safer traffic environment in the future. With full autonomy, the driver also has the possibility to do something else with the time otherwise spent driving. For the society, this could bring both societal and economic benefits. Since the driver is out of the loop, the vehicle has to be able to handle all possible situations that can occur. The result is a very large scope that has extreme dependability requirements related to safety. A large part of the focus right now is to make the vehicles capable of handling this large scope. Less is done on how to make sure that the dependability requirements are fulfilled. This is expected to be one of the greatest challenges in the commercialization of autonomous vehicles. To be able to overcome this, new methods have to be developed that can deal with a very large scope of requirements in a fast pace development. The safety has to be

quantified and it has to be shown with confidence that the vehicle is safe.

To validate complete vehicle safety, there are several approaches that have been applied. One approach is to test worst-case scenarios and assume that the less severe situations are handled if the worst-case is handled, e.g. [1]. This can be done for collision-avoidance systems by either using directed testing on a test course or by using computer simulations. The scope for collisions avoidance functions is narrow and well defined which means that worst-case situations are relatively simple to define. The challenge for autonomous vehicles is that the scope is much larger and less well defined, which makes the worst-case situations very difficult to identify. Since the scope is much larger, the number of test cases will also be much higher than for collision avoidance because it needs to cover all situations that can occur within the scope.

A second approach is to use models to verify the different parts of the autonomous vehicle work together safely. This can be realized in different ways. One method is to use stochastic simulations as described in e.g. [2] in order to explore the total space of possible scenarios. Another one is to use model based verification as in e.g. [3] and [4] to formally show that the system fulfills certain safety requirements. To validate system safety in this way requires valid models for everything that can affect the system. This includes how physical objects interact as well as the human behavior of drivers. To ensure a high validity of these models, an extensive amount of data from the traffic environment is needed.

A third approach is to use field tests and from that show statistically that the vehicle is safe enough, as described in e.g. [5]. This approach has been used to verify the safety of collision avoidance functions when it comes to false interventions. The difference is that while a collision avoidance function is passive most of the time, an autonomous driving function is active most of the time. This means that the safety requirement for an autonomous vehicle is much larger compared to a collision avoidance function. The result of this is that it would require covering extensive distances in various environments to show statistically that the system is safe.

The advantage of using field tests as a validation method is that the test environment has high validity. The situations are also directly sampled from the scope of the function. One issue is that collisions occur with such low frequency that we have to drive extreme distances in order to make a precise statistical estimate. New vehicles are often equipped with different sensors such as cameras and radars, which has access to data about its

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surrounding that is updated several times per second. This data can be used to calculate how close the vehicle is to a collision by using a threat measure. The frequency of events that are close to a collision can then be used for extrapolation to estimate the frequency of a collision. This has been done, using an area of statistics called Extreme Value Theory (EVT), in e.g. [6], [7], [8], [9] and [10]. The benefit of using closeness to a collision or near-collisions is that more of the available data can be utilized to make the statistical estimate, which leads to less variance of the estimation.

This paper is an extended version of [10]. The contribution of this article was to compare the usage of two different types of threat measure when using EVT. The data was gathered from a subset of a larger field test and the results suggested that one of the measures was more robust at predicting collision frequencies.

The contribution of this paper is to validate how well the collision frequency of vehicles can be estimated using EVT for different types of threat measures. The data set is a much larger set compared to [10], while the threat measures used are the same. The additions made in this paper is firstly an evaluation of multiple methods to automatically choose a suitable threshold, which is necessary in order to automate the process. Secondly, there is also a deeper analysis of the two different measures and reasoning about why they perform differently, connecting to previous studies. The main reason for using data from human drivers is that there exists a precise reference for the collision frequency based on crash statistics. Another reason is that there exist a significant amount of data in order to be able to draw statistical inferences related to the collision frequency.

In the following chapter, a background of relevant frequency statistics are presented. This is followed by a presentation of related works within traffic safety and the usage of EVT. Then the method of estimating the collision frequency from driving data using EVT is presented. In the results chapter, the results from a larger field test are presented for two types of near-collision measures together with some meta data. Then these results are analyzed in the discussion chapter by comparing the inferences drawn from the two types of measures. Finally, conclusions are drawn on how well these measures act as a near-collision measure and which questions that still need an answer.

II. BACKGROUND

In order to understand the magnitude of the statistical problem, knowledge about the frequencies of collisions is needed. Section II-A presents crash statistics of rear-end collisions, which is one of the most common types of traffic accidents. Then, two methods of modeling frequency data are described. The first method, presented in Section II-B, is using Poisson statistics to estimate the frequency of random events. The second method, shown in Section II-C, utilizes EVT to extrapolate near-crashes for estimation of collision frequency.

A. Crash Statistics

In Germany, crash statistics are well documented and it is one of the few countries that have documented statistics on

collisions with only property damage. A large part of the data set used in this paper is gathered in Germany as well. During 2014 in Germany, a total of around 740 billion km was driven according to [11]. During the same period 2.4 million accidents happened. In [12], it is stated that rear-end collisions account for almost 16% of the accidents in Germany. Assuming that 50% of the rear-end collisions are caused by the own vehicle, meaning that it is two vehicles involved in every collision, the distance between collisions will be 3.85 million km.

In the United States, over 3 trillion miles were driven during the same year according to [13]. During the same time around 6 million collisions were reported and out of these collisions around 32% were rear-end collisions, [14]. However, a large fraction of the collisions are not reported and studies such as [15] state that it is around 55%, while [16] found that it is around 29%. Using this data and the same assumptions as for the German statistics, the distance between rear-end collision is then between 2.2 and 3.5 million km. This supports that the estimate from the German accident statistics is reasonable.

B. Using Poisson Theory

The question that is being addressed here is how to make sure that failures leading to a collision are rare enough. This can be dealt with in the same way whether the failure is due to some error in an autonomous vehicle or if it is an error made by a human. A common method to deal with the frequency of random failure events is to treat them as a Poisson process [5]. In ISO 26262-8, [17], this is referred to under the concept of proven-in-use. The number of collisions during a certain time follows a Poisson distribution, which means that the time between collisions is exponentially distributed, $X \sim \exp(\lambda)$. We illustrate Poisson theory by an example:

Example 1: Suppose there is a requirement of a mean distance, μ , between collisions of 3.85 million km, i.e. the distance from German crash statistics. How far does a vehicle need to be driven, x , without a collision to be confident, with a certain level, $1 - \alpha$, that the mean is larger than the requirement? To test this, the null hypothesis is set to $\mu \leq 3.85$ million km, which is the same as $\lambda \geq \frac{1}{3.85} \times 10^{-6}$, since $\mu = \frac{1}{\lambda}$.

To reject the null hypothesis, the probability of getting a value x , or larger, has to be less than a certain risk level, $\alpha = 0.05$, for all values of $\lambda \geq \frac{1}{3.85} \times 10^{-6}$. The 95th percentile of the cumulative distribution function, F , can be used to find this value of x .

$$F(x; \lambda \mid x \geq 0) = 1 - e^{-\lambda x} \Rightarrow 1 - e^{-\lambda x} = 0.95 \quad (1)$$

For a larger λ , the risk level α will be smaller, which means that we only need to consider the case where $\lambda = \frac{1}{3.85} \times 10^{-6}$. Putting this value for λ into (1) and solving for x result in:

$$x = \frac{-\ln(0.05)}{\frac{1}{3.85} \times 10^{-6}} \approx \frac{3}{\frac{1}{3.85}} \times 10^6 = 1.16 \times 10^7. \quad (2)$$

This means that if 11.6 million km has been traveled without an accident, we can reject the null hypothesis that $\mu \leq 3.85$ million km with a confidence of 95%.

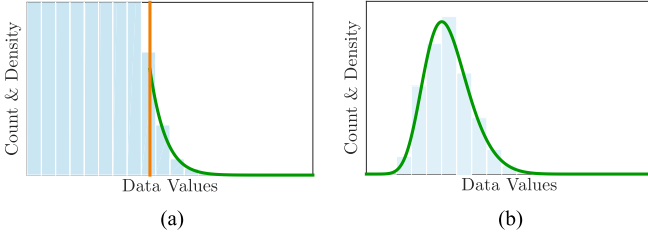


Fig. 1. Illustration of two different methods of modelling extreme values. (a) Peak over Threshold method. (b) Block Maxima method.

Poisson statistics only uses the actual collisions as data. With vehicles now being equipped with sensors that can tell the closeness to a collision, more data can be used for the statistical estimation. One area of statistics that can be used with this type of data is EVT.

C. Using Extreme Value Theory

In EVT, extreme events are modeled using a statistical distribution, which then can be used for extrapolation.

There are two main procedures in EVT of how to do the modeling, which is illustrated in Fig. 1. The first method is called Peak over Threshold (POT) and the second is called Block Maxima (BM), [18]. Both distributions consist of three parameters; shape (ξ), scale (σ) and location (μ). The density of more extreme values from the distributions can be used to estimate frequencies of events that have not yet occurred.

In the POT method, local maxima of the data are sampled and the values over a certain threshold are fitted to a Generalized Pareto (GP) distribution. This is shown as the green solid line in Fig. 1(a), which density function is

$$f(x|\xi, \sigma, \mu) = \frac{1}{\sigma} \left(1 + \xi \frac{x - \mu}{\sigma} \right)^{-(1/\xi + 1)}. \quad (3)$$

The BM method instead splits the sampling procedure into blocks of a certain time length and samples the maximum value in each block. The block maxima are then fitted to a Generalized Extreme Value distribution, which density function is

$$f(x|\xi, \sigma, \mu) = \frac{1}{\sigma} \exp \left(- \left(1 + \xi \frac{(x - \mu)}{\sigma} \right)^{-\frac{1}{\xi}} \right) \times \left(1 + \xi \frac{(x - \mu)}{\sigma} \right)^{-1 - \frac{1}{\xi}}. \quad (4)$$

III. RELATED WORK

EVT has been used previously to estimate collision frequencies using near-crashes as data. In most of these studies, time-based measures have been used to represent the closeness to a collision. In [6], intersections were studied using cameras and collision frequency was estimated using EVT. The estimate was compared with Poisson confidence intervals from crash statistics. However, due to short observation time, the estimates varied a lot and the result was difficult to validate.

In [9], EVT and the BM method was used to estimate the frequency of road departures on a specific road segment. Data

was gathered using e.g. GPS, camera, radar and a detailed map to calculate Time to Edge Crossing (TTEC) that was used as a near-crash measure. The result suggested that a road departure would occur about 12 times a year. This was compared to the 1.8 crashes that occur due to road departures at this road segment. The conclusion was that the estimate was reasonable and highlighted the value of future more in-depth research. In [7], data is gathered from a driving simulator and Time to Road Departure is used as a near-crash measure. The collision frequency is estimated using EVT and the POT method is used instead of the BM method. Common for these studies is that the data sets used have been relatively small and the results can only be seen as indicative.

In [8], vehicle-mounted forward-looking radar data from the 100-car study, a large naturalistic driving study, is used as input. The data was gathered using different kinematic triggers containing combinations of acceleration and Time to Collision (TTC), summarized in [19]. This resulted in 384 near-collisions and 14 collisions that were classified as rear-end. TTC was used as a measure to the closeness of a collision and the BM method was used to estimate the rear-end collision frequency. This estimate was compared with a binomial estimate from the actual collisions. The result was that the EVT estimation of collision frequency was 175 times lower than the actual collisions. One of the main concerns from the results was the substantial internal selection bias. Almost all collisions were in slow-moving traffic, while the near-collisions were in free-flowing traffic. Inconsistent radar data also led to only 29 usable near-crashes, which is mentioned as one of the factors for the discrepancy. Even though the data set is large, the quality of the data makes it difficult to draw conclusions about the applicability of the method.

In [20], TTC is also used as threat measure, but in this case for head-on collisions during overtaking maneuvers. The data comes from experiments in a simulator of overtaking situations with varying parameters. Almost 1300 maneuvers were performed containing 9 collisions and the TTC for the non-collisions were fitted using the Block Maxima method to estimate the collision frequency. The result was that the EVT estimate was close to the actual collisions. In [21], both the BM and POT methods were applied to a similar scenario. There, covariates were included in the model to compensate for the speed bias of TTC with successful results.

Instead of having a measure that relates to the closeness to the point of collision, like TTC, there are measures that measure the closeness to a point where a collision is unavoidable. In [10], two types of near-crash measures are compared using EVT to estimate collision frequency, TTC and Brake Threat Number (BTN). The results show that BTN, measuring the closeness to a point where a collision is unavoidable by braking, was the more robust type.

IV. METHOD

This section describes the process of using logged data of the surroundings in order to estimate the distance between collisions for vehicles. A threat measure assesses the closeness to have a collision based on the information about the surroundings

gained from logged data. In this case, for rear-end collisions, the threat measures used are BTN and TTC. The most extreme cases of these measures are then modeled using EVT in order to estimate the distance between collisions. The confidence interval of the estimate can then be used to validate that the true distance between collisions is above a certain requirement with some confidence.

A. Logged Data

There is usually a trade-off between getting a large amount of data and getting data with high quality. This is because good sensors are expensive, and usually not fitted to production vehicles, which limits the ability to do large scale data collections. For our purpose, it is important to have a sample as large as possible. At the same time, the quality of the sensor data needs to be good enough to not give a significant impact on the results. A forward-looking radar and camera based system fit this purpose for rear-end collisions. The raw data from the sensors is processed and fused into objects with different properties associated with them. Also, information about lane markings is available, which is used to create a map of the road. The data is captured with a frequency of 40 Hz, which means that the stream of observations is continuous. This makes POT the better choice of EVT method due to that less data is thrown away as stated in [18]. To only include situations where the ego vehicle is on collision course, only objects that are in the same lane are considered. To make a prediction of the objects' movements, constant relative lateral speed and constant relative longitudinal acceleration are assumed. If the ego vehicle is predicted to collide with the object, it is included in the analysis.

B. Threat Measure

To fit the EVT model to the data, a measure that reflects the seriousness of the situations is needed. There are several types of threat measures. TTC and BTN, which are used here, are deterministic because they use a single trajectory as the prediction for each object [22]. There is also stochastic threat assessment, which considers multiple trajectories that are weighted based on a stochastic model of the behavior. A probability of collision for each state can then be calculated, e.g. [23] and [24]. These stochastic models often rely on the input of the type of traffic scenario or visibility, which is not available in this data set. The scope of rear-end collisions with other vehicles also has a limited number of possible actions, which makes the prediction simpler. Using stochastic threat assessment requires more computing, which is a limiting factor when dealing with large data sets such as this. In this paper, the focus is therefore on the deterministic threat assessments, but for more complex traffic scenarios, stochastic threat assessment could lead to better results, as mentioned in [25].

For the objects that are left after the filtering, the threat measures BTN and TTC are calculated in each time frame. TTC is calculated assuming constant acceleration based on the following equation:

$$\frac{\ddot{x}_0 t^2}{2} + \dot{x}_0 t - x_0 = 0, \quad (5)$$

where $\ddot{x}_0$ is the relative longitudinal acceleration and $\dot{x}_0$ is the relative longitudinal velocity between host and object at the current time. The TTC value is received by solving the equation with regards to the time, t , and choosing the lowest positive root. If there is no positive root TTC is infinite.

BTN is the relation between the negative acceleration needed to marginally avoid a collision and the maximum deceleration available for the vehicle. If the object is non-closing, i.e. the required acceleration is positive, the BTN is set to zero. The required acceleration can be calculated using the following two equations for the relative velocity and position after the time t :

$$\dot{x}(t) = \dot{x}_0 + (a_{obj} - a_{req})t = 0 \quad (6)$$

$$x(t) = x_0 + \dot{x}_0 t + (a_{obj} - a_{req}) \frac{t^2}{2} = 0, \quad (7)$$

where a_{obj} is the object's current acceleration and a_{req} is the required acceleration. Inserting the expression for t from equation 6 into equation 7 and solving for a_{req} will yield:

$$a_{req} = a_{obj} - \frac{\dot{x}_0^2}{2x_0} \quad (8)$$

$$BTN = \frac{a_{req}}{a_{max}}. \quad (9)$$

The maximum deceleration under full braking is in this case assumed to be -9.82 m/s^2 [22].

The most threatening object for each time frame, the one with the most extreme threat value, is selected and this results in one time series of threat values for each measure. From these time series, peak values are extracted with a minimum of 30 seconds separation. This is because the POT method requires independent observations and this time interval is long enough to not sample from the same extreme situation. At the same time, not too much data is thrown away. After the extraction of peaks, the most extreme peaks are investigated by video to ensure that the sensor data correctly represents the situation. This is done because these peaks have a large impact on the EVT extrapolation.

C. Extreme Value Theory

To model the threat measure values using EVT and POT, a threshold over which the values are regarded as extreme has to be determined. This is done by investigating the stability of the fitted distribution for different thresholds. For each possible threshold, the exceeding values are subtracted with the threshold value to form what is called the exceedances. The exceedance data is fitted to the GP distribution, using maximum likelihood estimation, as described in [26], and values for scale, σ , and shape, ξ , are received for different thresholds. In order for the scale parameter to be comparable with different thresholds, it has to be reparameterized with regard to the threshold, u , and the shape parameter:

$$\sigma^* = \sigma_u + \xi u. \quad (10)$$

When both σ^* and ξ are constant, the estimation is stable during that interval. The appropriate threshold to choose is the least extreme of the thresholds, where all the more extreme thresholds

follow the same distribution [18]. The reason for this is that the data is EVT distributed while as much data as possible is used, which reduces the uncertainty of the extrapolation.

To choose this threshold in practice is not trivial. This is often done by inspecting parameters and other indicators visually, which requires a lot of experience and may also be subjective. In recent years some automated threshold selection algorithms have been developed. This is required in order to make batch estimations or other automated analyses using EVT.

In this study, three different methods of finding a stable threshold are used. The first two are presented in [27]. Both versions estimate a threshold by determining how many of the upper extremes, k , that should be used for the estimation. Let $Z_i = X_{n-k+i} - X_{n-k}$ be the ordered exceedances, where n is the sample size, and let ξ_k be the estimate of the shape parameter for $Z_1, \dots, Z_k$. The problem is then to find the k which minimizes the total deviation

$$D(k) = \frac{1}{k} \sum_{i \leq k} i^\beta e(i, k), \quad (11)$$

where β is a scaling factor, i is the number of extremes used for estimation, and $e(i, k)$ is the respective deviation error. The scaling factor β is selected somewhere between 0 and 0.5. This controls how the weight is put on errors for different number of extremes.

The first method presented in [27] is using the absolute deviations of ξ_i from the median as the deviation error, $e(i, k) = |\xi_i - \text{med}(\xi_1, \dots, \xi_k)|$. This algorithm will hereby be referred to as method A.

The second method presented in [27] is instead using squared deviations relative to ξ_k , $e(i, k) = (\xi_i - \xi_k)^2$. This algorithm will hereby be referred to as method B.

The third method used is presented in [28], which calculates a discrepancy measure that should be minimized. This method will hereby be referred to as method C. It is based on the assumption that the CDF value of each exceedance, $F_{\hat{\sigma}, \hat{\xi}}(Z_i)$, should be uniformly distributed. The discrepancy measure, $D(k)$, for the minimization problem is stated as

$$D(k) = \frac{1}{k} \sum_{i=1}^k \left[\frac{F_{\hat{\sigma}, \hat{\xi}}(Z_i) - i}{k+1} \right]^2. \quad (12)$$

In (12) the average squared deviations from the expected uniform distribution is evaluated for each value of k . For all three methods, the best k is the one that minimizes $D(k)$ and the threshold is then chosen as X_{n-k} .

There is often the case in applications that there are several possible thresholds, [29]. These algorithms can, therefore, find different threshold depending on what the weight is put on. It is important that the sensitivity of the inferences drawn is evaluated for different possible thresholds.

The distributions from the possible thresholds can be used to estimate the distance between collisions, as described in [10]. Related confidence intervals can also be calculated using a profile of the log-likelihood. In this paper, a concept called return level, which is the most extreme value to expect after a certain period, is used to visualize this.

A certain distance traveled, m , leads to a number of peaks exceeding the threshold, k_m , which estimate is equal to

$$\hat{k}_m = k \frac{m}{m_{tot}}, \quad (13)$$

where k is the total number of exceeding peaks and m_{tot} is the total distance. The estimate of the quantile, p , of the GP distribution corresponding to the distance m is

$$\hat{p} = 1 - \frac{1}{\hat{k}_m}. \quad (14)$$

A confidence interval from the estimation in (13) can be used to give a span of possible quantiles. Given the maximum likelihood estimates of ξ and σ , the return level, x_m , is given by the quantile of the inverse cumulative distribution function:

$$x_m = F^{-1}(\hat{p} | \hat{\xi}, \hat{\sigma}) + u. \quad (15)$$

To find the confidence interval of the return level, profile likelihood intervals are used. These have better accuracy of the uncertainty for the extrapolation than the delta method according to [18]. The likelihood for different values of ξ and σ is calculated to create a surface profile of the likelihood. Then a lower log-likelihood limit, $\log \mathcal{L}_\alpha$, is calculated based on the maximum likelihood value and the χ^2 -distribution:

$$\log \mathcal{L}_\alpha = \log \mathcal{L}(\hat{\xi}, \hat{\sigma}) + \chi_{1-\alpha, 1}^2. \quad (16)$$

The parameters ξ and σ are chosen so that x_m is minimized or maximized while having a higher likelihood than the lower log-likelihood limit for a certain risk-level, α . This gives the highest or lowest probable value of the threat measure after a certain distance driven.

V. RESULTS

In this section result from field test data will be presented. The data consist of around 250 000 km of driving done by test drivers in a mixed driving environment. One of the reasons for using data from human drivers is that the estimations can be compared with reference from crash statistics. Data collection was made in Europe with a focus in Germany and Sweden. This has resulted in around 130 000 peak values of BTN and around 140 000 peak values for TTC. To ensure the validity of the data, the 600 most extreme peaks has been investigated for each of the two measures using both video and sensor data visualization. After this point, the ratio of invalid peaks is very low and also have little impact on the estimation.

A. Using Brake Threat Number

The first step of the POT method explained in the method section is to find a valid threshold. This has, in this case, been identified using the different methods presented in IV-C. The results from these methods, shown in Table I, indicates several possible thresholds, which needs to be investigated separately as described in the same subsection.

The first threshold seen from the right side in Fig. 2 has a value of around 0.337 and is selected by method A. The shape estimations for higher thresholds are similar, which indicates a

TABLE I
THE RESULT OF USING BTN AS A MEASURE

Method	k	u	$\hat{\xi}$	ξ_{CI}
A	176	0.337	-0.0852	0.0508
B	5429	0.170	0.102	0.126
C	3477	0.190	0.130	0.162

For each method, the threshold, u , and the number of peaks above the threshold, k , are presented. Both the maximum likelihood estimate and the lower confidence estimate for the shape parameter, ξ , are also shown for each method.

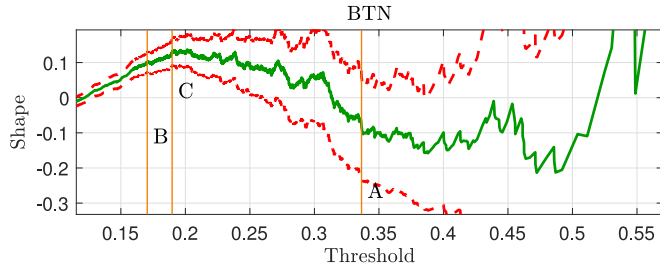


Fig. 2. Shape parameter estimations for different thresholds using BTN as threat measure. The green line is the maximum likelihood estimate and the dashed red lines represent a 95% confidence interval. The thresholds from the three different methods are marked with orange lines.

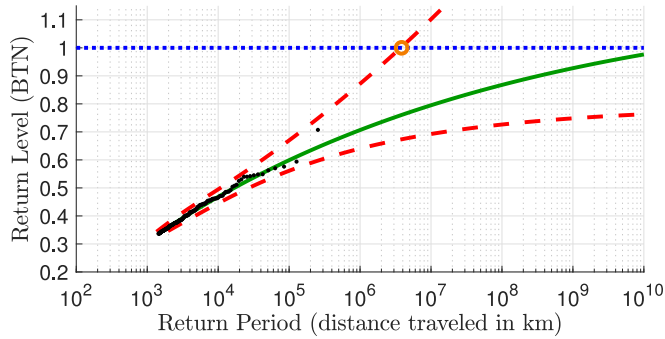


Fig. 3. Return Level plot for BTN using method A to select threshold. The green line is the maximum likelihood estimate and the red dashed lines represent a 90% confidence interval. The black dots along the estimation represent the exceedance data and the dotted blue line is the critical limit. A reference to the actual distance between collisions, described in Section II-A, is marked with an orange circle at the critical limit.

good fit of the data. The other two selected thresholds are relatively close to each other but differ a little in the estimated shape parameter. The lower threshold is around 0.17 and selected by method B and the higher threshold is around 0.19 and selected by method C. There is also a notable large shift in the shape parameter when going from the higher threshold selected from method A to these lower thresholds. This indicates a significant difference between the estimated distributions. However, they are both close to a plateau of relatively stable parameter estimation. The decreasing shape for higher thresholds is explained by the increased share of the more extreme data that is reflected in the shape of threshold selection A.

Fig. 3 shows the concept of return level explained in sub-Section IV-C. The intersection between the estimation line and the dotted critical limit line is the estimated distance between

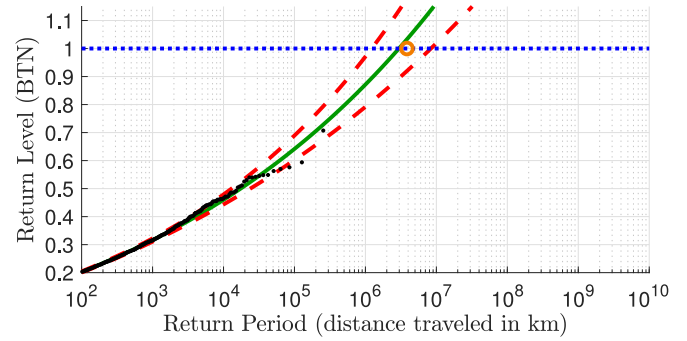


Fig. 4. Return level plot using method B to select threshold. The lines have the same representation as in Fig. 3.

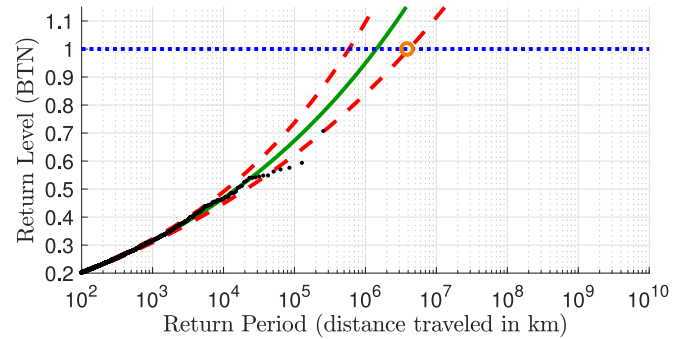


Fig. 5. Return level plot using method C to select threshold. The lines have the same representation as in Fig. 3.

collision from the data. In this case, only the data from the outer part of the tail is used to fit the distribution. A little more than 0.1% of the total number of peaks are used in the estimation. The data follows the distribution well for the most part. There is a notable deviation around 0.55 where there are several values close to each other. The estimated distance between collisions is far away from the estimate from crash statistics. However, the lower limit of the 90% confidence interval is still lower than this estimate.

In Fig. 4, the estimate is close to the reference from crash statistics, which is well within the 90% confidence interval. The data fits the distribution well for low threat values. However, BTN values above 0.4 are deviating somewhat from the estimated distribution. This can also be seen in Fig. 2, with the shape estimation getting lower with larger thresholds. There are several higher BTN values that are very close to each other, which is also seen in Fig. 3. This results in the deviating parameter value estimations seen in Fig. 2, which affects especially higher threshold values.

Fig. 5 shows the estimation resulting in the shortest distance between collisions with a threshold. The estimated distance between collisions is lower than the crash statistics estimate. However, the crash statistics estimate is still within the 90% confidence interval. A result of this lower estimation is that some of the most extreme data points are outside of the confidence bounds, which indicates a bad fit for the end of the tail. The other part of the data follows the estimation relatively well and the deviations after 0.4 are less pronounced compared to Fig. 4.

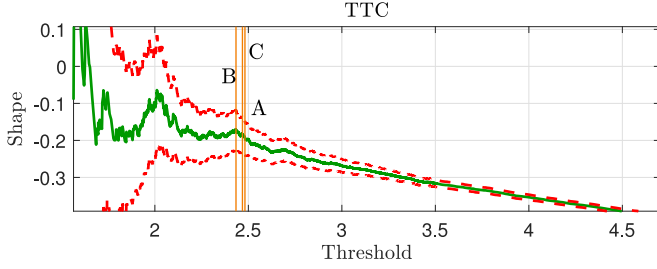


Fig. 6. Shape parameter estimations for different thresholds using TTC as threat measure. The lines have the same representation as in Fig. 2.

TABLE II
THE RESULT OF USING TTC AS A MEASURE

Method	k	u	$\hat{\xi}$	ξ_{CI}
A	1104	2.48	-0.193	-0.152
B	974	2.43	-0.170	-0.123
C	1073	2.47	-0.183	-0.140

The table presents the same parameters as Table I.

B. Using Time to Collision

Using TTC as a threat measure instead, the first thing that changes is that lower numbers are the more extreme situations. This means that the axis will be mirrored, which can be seen in the stability plot in Fig. 6. In the return level figures, the TTC values are shown as negative in order to have the highest values reflecting the most extreme events. This makes it easier to compare the figures with the ones using BTN as a threat measure. For TTC the critical level is 0 since that means that a collision has just happened. This is shown by the blue dotted line in the same way as with BTN.

In Fig. 6, there is clear indication that thresholds above 2.5 are not extreme value distributed. Compared to BTN, a much lower share of the data seems to be EVT distributed. In both of the figures for threshold selection the maximum threshold on the x-axis represent 12.5% of the data. This means that the lower thresholds for BTN are using many times the amount of data compared to TTC, as seen when comparing the k -values from Tables I and II. Lower thresholds lead to a large flickering of the parameter estimations. In this case, all methods result in thresholds close to each other and just below 2.5, where the significant shift happens. Method A selects the highest threshold of 2.48 while method B selects the lowest at 2.43. This is the same order of the threshold methods as for BTN. Even though the thresholds are very close to each other, both the highest and the lowest will be shown to highlight the differences that the choice results in.

The estimation in Fig. 7 is based on the highest of the selected thresholds. Therefore, a little bit more data is included and the estimation is more biased toward the less extreme values. This can be seen by the most extreme data points being close to, or even outside, the confidence interval in the return level plot. Also, the most extreme values are very close to each other, which leads to a lower shape parameter. The result of this is that all the estimations, even the lower limit, is never crossing the critical

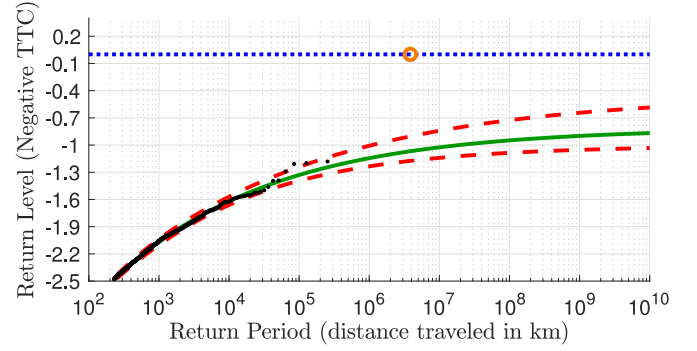


Fig. 7. Return level plot using method A to select threshold. The lines have the same representation as in Fig. 3.

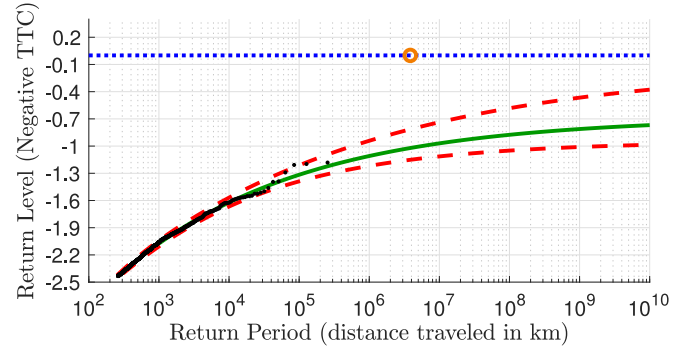


Fig. 8. Return level plot using method B to select threshold. The lines have the same representation as in Fig. 3.

limit. This suggests that the vehicles will never make a rear-end collision.

The lowest threshold includes a little bit less data, which makes the estimation less certain. This can be seen in Fig. 8 as a wider confidence interval around the estimation. The result of this is that the lower limit is closer to the critical limit. A higher value of the shape parameter estimation also helps to push the estimations a little bit higher towards the more extreme values. However, the estimation still suggests that with a very high confidence that a collision will never occur.

C. Speed Distribution

To gain some more knowledge about what is selected as the most extreme situations, we can look at the ego speed distribution. This will give an indication of what type of traffic situation that is more present depending on the chosen threat measure.

The distribution of TTC is highly biased towards low-speed situations, which can be seen in Fig. 9. Most of the events occur with an ego vehicle speed of less than 10 m/s and almost no representation of high-speed situations. If comparing to the total population, there is evidence of over-representation of low-speed situations.

For BTN, the situation is the opposite. Almost none of the most extreme situations occur with an ego vehicle speed of less than 10 m/s. The values are also more evenly spread compared to TTC with a large representation of high-speed situations. Comparing BTN with the total population, the events are more evenly sampled. The low-speed situations are undersampled, while the

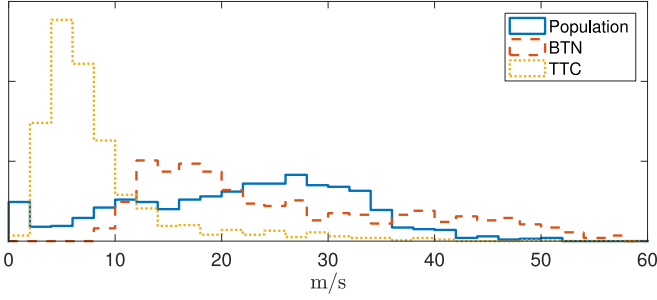


Fig. 9. Histogram showing the ego vehicle speed for the 1000 most extreme values of TTC and BTN respectively. As a reference, the distribution of ego vehicle speed for the whole data set is shown and scaled to match the other two histograms.

high-speed situations are somewhat oversampled. Otherwise, the density is relatively similar with some characteristics found in both.

D. Comparison with Poisson

The inferences drawn from Fig. 3 is that there is a 95% probability that the distance between collisions is more than 3.74 million km. If the same statistical inference should be drawn using the method in Example 1, the vehicle would need to be driven 11.2 million km without an accident. This is 45 times longer distance than what is driven in the data set used in this paper.

To compare the confidence of the estimations, the estimation is seen in Fig. 4 can be used. It has an estimate of 3 million km between collisions and 90% confidence interval of [1.22, 8.38] million km. To get within the same limits with the same average frequency using Poisson statistics one would have to drive 15 million km and have 5 collisions. This is 59 times more driving data than used for EVT. The corresponding exact confidence interval for the collision frequency would, in that case, be [1.41, 7.53] million km, as applied in e.g. [30].

VI. DISCUSSION

The following discussion will be centered around three different areas. Firstly, the results from the two threat measures will be discussed and why the estimations differ from each other. Then, a reflection will be made on how these results relate to results from previous work, presented in the literature. Lastly, there is a discussion about how the data from human driven vehicles could differ from autonomous vehicles and what implications that will have for using this method.

A. Threat Measures

The choice of the threshold of a threat measure has a large impact on the inferences. For BTN, the best choice of threshold is not very clear. Every method chooses a different one and there is a big difference in the inferences drawn from them. The higher thresholds suggest a distribution with a lower shape parameter while lower thresholds result in a higher. There seem to be two different distributions present in the tail data. A difference in

human behavior between situations of different severity could be one explanation to this.

The estimation in Fig. 3, using 0.34 as the threshold, results in a shape which suggests that the vehicle will crash very rarely. This is due to that several of the most extreme peak values are relatively close to each other. There seems to be some dampening effect that stops the BTN values of getting too large. This might have something to do with that there are humans driving, which do not want to get too close to a collision. The fact that the vehicles are driven by trained and rested test drivers, should lead to a better than average collision frequency. The data points also follow the estimation rather well, which supports the correctness of this fit. The crash statistics estimate is still within the confidence interval, but the uncertainty is relatively high due to that few data points are used.

Using threshold 0.17 or 0.19, as shown in Figs. 4 and 5, result in very similar estimations. The lower threshold results in an estimation very close to the crash statistics estimate, which could indicate that this is a good threshold. However, the lack of fit for the more extreme data point adds doubt to the estimations from these thresholds. It is more biased towards normal driving conditions with a more moderate threat. Using a lower threshold seems to have a tendency to overestimate the frequency of a collision. From a safety validation point of view, this is better than underestimation, but it might require more data to validate the same requirement. The confidence intervals for these thresholds are narrow because more of the data is used for extrapolation.

For TTC, the situation is different. All the methods choose almost the same threshold, giving a strong indication that it is a suitable one. However, there is a large variance of the shape parameter estimation for the more extreme thresholds, which adds some uncertainty. The result from using the thresholds for TTC is an estimation, which suggests that there will never be a collision. Even the lower limit never crosses the critical limit, as seen in Fig. 8. The shape parameter of the TTC distribution is negative compared to a positive shape of the BTN distribution. The most extreme data points for TTC are also very close to each other, which contributes to a more negative shape of the distribution. These results are very similar to the results from a smaller field test in [10] and further strengthens the conclusions made in that study.

B. Relation to Previous Work

In articles based on the 100-cars study such as [8], one of the main concern was the substantial internal selection bias. In the data set used in this article, the ego speed is low for the most extreme situations of TTC, as seen in Fig. 9. This is consistent with the fact that most actual collisions in the 100-car study happened in slow-moving traffic. These types of collisions are often due to inattentiveness, which is less likely to be present with the trained drivers in this field test, speaking against that the most extreme situations are low-speed situations. An explanation to this is that TTC, as a measure, is highly dependent on the speed. For example, a low TTC in slow-moving traffic might not be such a critical situation because the stopping time is very short. At a high speed, the same TTC could be a

very critical situation that often leads to collisions. The result from this is that two situations with the same TTC might not have the same probability to lead to a collision, which is not wanted when using EVT. This choice of threat measure has not been discussed as a possible cause of result discrepancy. The dependence of speed could also be an explanation of the big difference when comparing the speed distribution of TTC with the distribution for the whole data set, Fig. 9. The estimation from this data set using TTC similarly ends up with a collision frequency, which is much lower than the actual one. In [21], this is compensated by including covariates related to speed. However, it is preferable to use a measure which is stationary from the beginning to avoid modifying the statistical model.

In [20] TTC is working well to predict the collision frequency. This is probably because head-on collisions during overtaking are different from rear-end collisions. In overtaking maneuvers, the overtaking vehicles have a certain time to finish the maneuver, which is directly linked to TTC. This means that the same value of TTC reflects similar conditions for the overtaking maneuver. For rear-end collisions situations this is not the case because of a large variation in the braking time, dependent on the relative speed between the vehicles. In the case of overtaking maneuvers, this dependence is much smaller.

C. Human Versus Autonomy

The different threshold levels for BTN suggest that there might be two different EVT distributions present. Using the highest threshold reflects the distribution of the most extreme events, where the vehicle is closer to a collision. The lower thresholds result in an estimation reflecting more the situations of moderate threat. Since there is a clear shift of the distribution, it suggests that there is a difference in the underlying behavior in these situations. While the moderate situations point to a collision frequency which is higher than average, the more extreme situations indicate a much lower collision frequency. The driver seems to be able to handle more critical situations better than what the more moderate situations suggest. For TTC there is no evidence in the data of this difference. One reason for this could be the difference in traffic situations for the most extreme events for TTC and BTN, shown by the distribution of ego vehicle speed in Fig. 9.

For autonomous vehicles, the difference in behavior for different threat levels will probably change. Some part of it could still exist if there is a different part of the software which handles the regular driving compared to the collision avoidance situations. However, the difference between these parts will probably not be as large as it is in this data. The autonomous system will act earlier, not be distracted and by that act more reflecting to the seriousness of the situation. The transition from regular driving to collision avoidance will probably also be smoother than for a human.

One important note to make is that this method can only account for causes of a collision that is visible in the threat measure. In this case, it means that small safety margins and inattentiveness should be visible, but that more rare causes, such as heart attacks, are probably not visible. The same thing

exists for autonomous cars, where errors such as late detection of objects are visible, as opposed to a failure of a single ECU in a redundant architecture. A difference in this regard is that we will know much more about the possible causes of errors in an autonomous car than what a human driver may do wrong. For an autonomous vehicle, it is possible to test and verify the error rates for many of these causes, while a human is more unpredictable and more difficult to measure.

VII. CONCLUSION

In this paper, a large data set containing over 250 000 km driving data has been used to estimate the collision frequency using EVT.

The result from this study further strengthens the conclusions drawn in [10] about using different types of threat measures for EVT. The choice of threat measure has a large impact on the inferences drawn from the same data. Using BTN as a threat measure results in estimations which confidence intervals all include the crash statistics estimate. The deviation of the estimate for one of the thresholds can be explained by the trained drivers, with better than average capabilities. In contrast, using TTC as a threat measure results in estimations which suggest that there will never be a collision. This highlights the importance of choosing a threat measure which is comparable for different types of traffic situations.

All three threshold selection methods choose probable threshold values. For BTN, the choices are split between the methods because of the difference in behavior for more severe situations. This could still exist for autonomous vehicles, but the variance in behavior will be much smaller since the same software will be driving the vehicle for all data collected. For autonomous vehicles, it will also be important to use a threat measure that reflects the vehicles limitations and not the ones of a human. TTC has often been used due to its natural connections to the human reaction time, while BTN instead is related to the vehicle's braking capabilities. The estimations for BTN in relation to the crash statistics estimate support that it is a suitable threat measure for predicting rear-end collisions using EVT.

In general, with the right measure, EVT can be used as a safety validation method. The validity of the data is kept high since it will be sampled from real traffic. It also uses the available data more efficiently compared to state of the art used statistical methods, Poisson analysis. EVT required 45 times less driving distance to draw the same inferences, which makes it possible to apply for the strict requirements of autonomous vehicles.

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